

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			bonds broken $3(\text{C—C}) + 9(\text{C—H}) + (\text{C—O}) + (\text{O—H}) + 6(\text{O=O})$ $1044 + 3708 + (\text{C—O}) + 463 + 2976$ $8191 + (\text{C—O})$ (1) bonds formed $8(\text{C=O}) + 10(\text{O—H})$ $6440 + 4630$ 11070 (1) enthalpy change $8191 + (\text{C—O}) - 11070 = -2676$ (1) $(\text{C—O}) = -2676 - 8191 + 11070$ $(\text{C—O}) = 203$ (1) ecf possible throughout						
				Question 12 total	2	8	1	11	6	4